FTK Imager – Forensic Image Creation Report

Practical Assignment: Logical Acquisition of a Test Folder

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2026-09-03

1. Student Information

Table 1 Student Information

Field	Details
Name	Lisa Kabue
Roll/Enrollment Number	FTK-001
Course/Subject	Digital Forensics
Assignment Title	Forensic Image Creation Using FTK Imager
Date	2026-09-03

2. Software Information

Table 1 Software Information

Item	Details
FTK Imager Name	Exterro FTK Imager
Version Installed	8.3.0.27
Operating System Used	Windows (HomePC)

3. Step-by-Step Procedure

3.1 Step 1 – Download FTK Imager

Navigated to the official Exterro download page at https://www.exterro.com/digital-forensics-software/ftk-imager and downloaded the FTK Imager installer. The download file was named **FTK Imager 8.3.0.27.zip** (387 MB). The browser download progress was monitored until the file was fully downloaded.

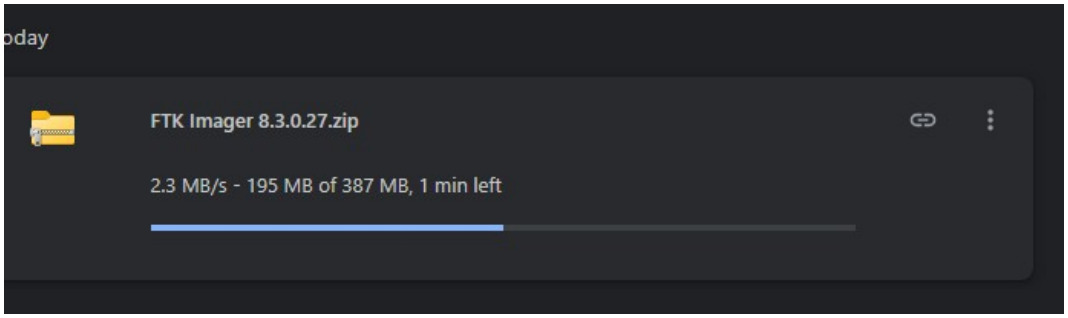


Figure 1 Downloading FTK Imager 8.3.0.27 from the official Exterro website

3.2 Step 2 – Install FTK Imager

The downloaded ZIP file was extracted, and the installer **Exterro_FTK_Imager_(x64).exe** (224,379 KB) was located inside the extracted folder along with an ISSetupPrerequisites subfolder. The installer was run with administrative privileges. The installation wizard guided through the license agreement, installation path selection, and component selection. The installation was completed successfully.

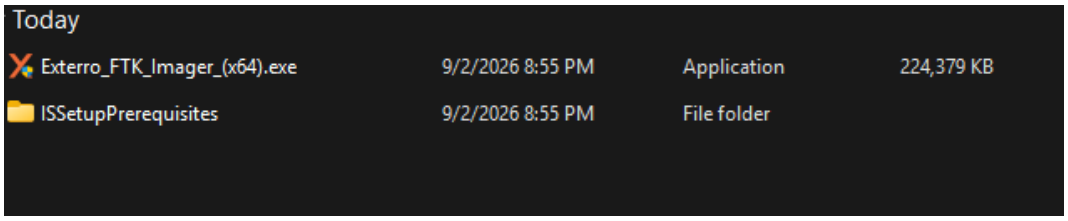


Figure 2 FTK Imager installer file located in the extracted download folder

3.3 Step 3 – Launch FTK Imager

After installation, FTK Imager was launched from the desktop shortcut or Start Menu. The main interface of **FTK Imager 8.3.0.27** appeared successfully, displaying the menu bar (File, View, Mode, Help, Upgrade to PRO), toolbar icons, and the Evidence Tree, File List, Custom Content Sources, and properties panels. The application opened without any errors, confirming a successful installation.

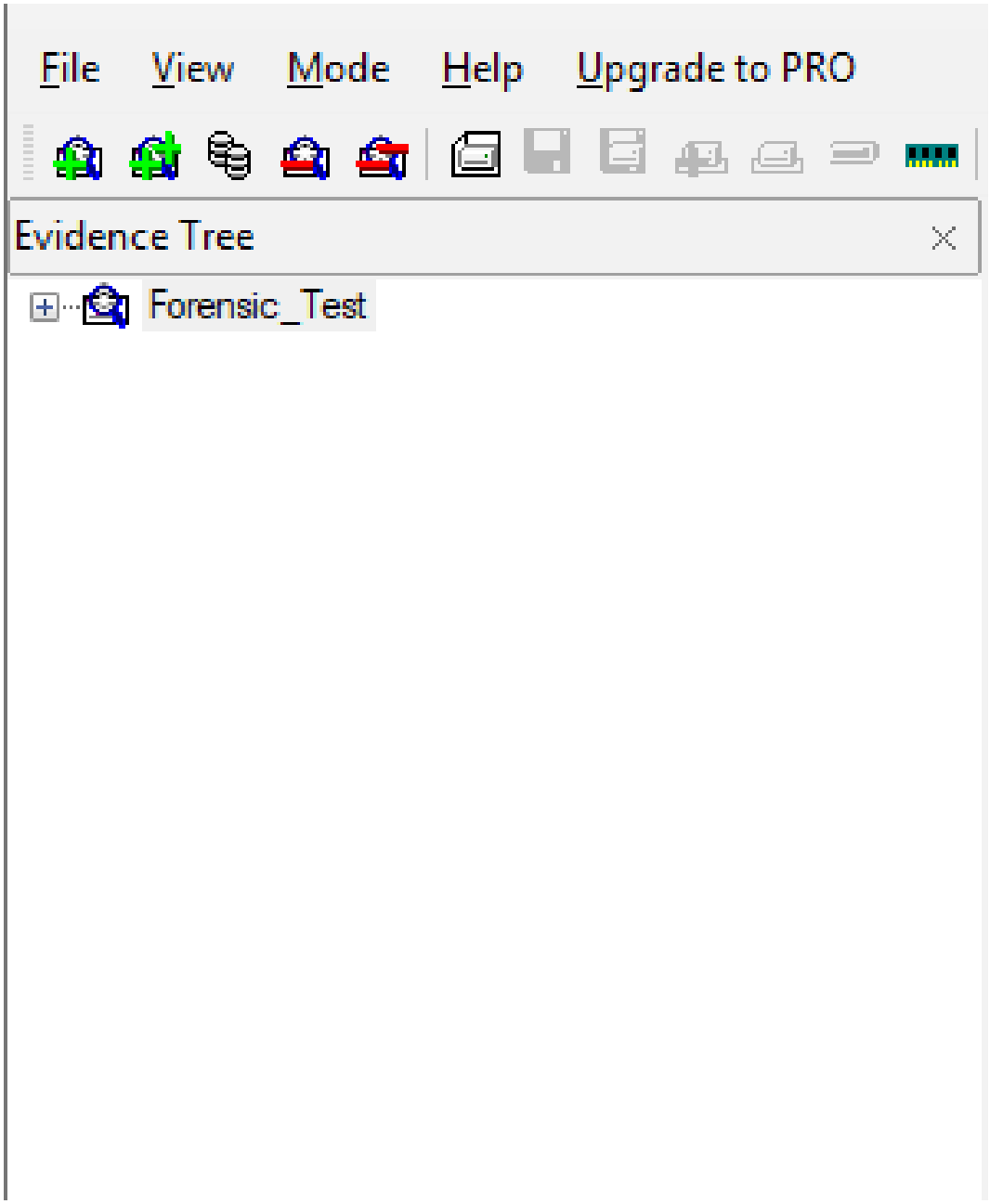


Figure 3 FTK Imager 8.3.0.27 successfully opened and ready for use

3.4 Step 4 – Select the Evidence Source

From the File menu, **File → Create Disk Image** was selected. The "Select Source" dialog appeared, prompting for the type of evidence source. The following options were available: Physical Drive, Logical Drive, Image File, and Contents of a Folder. Since this exercise involves a logical acquisition of a folder, **"Contents of a Folder"** was selected. The description noted: "logical file-level analysis only; excludes deleted, unallocated, etc." The "Next" button was clicked to proceed.

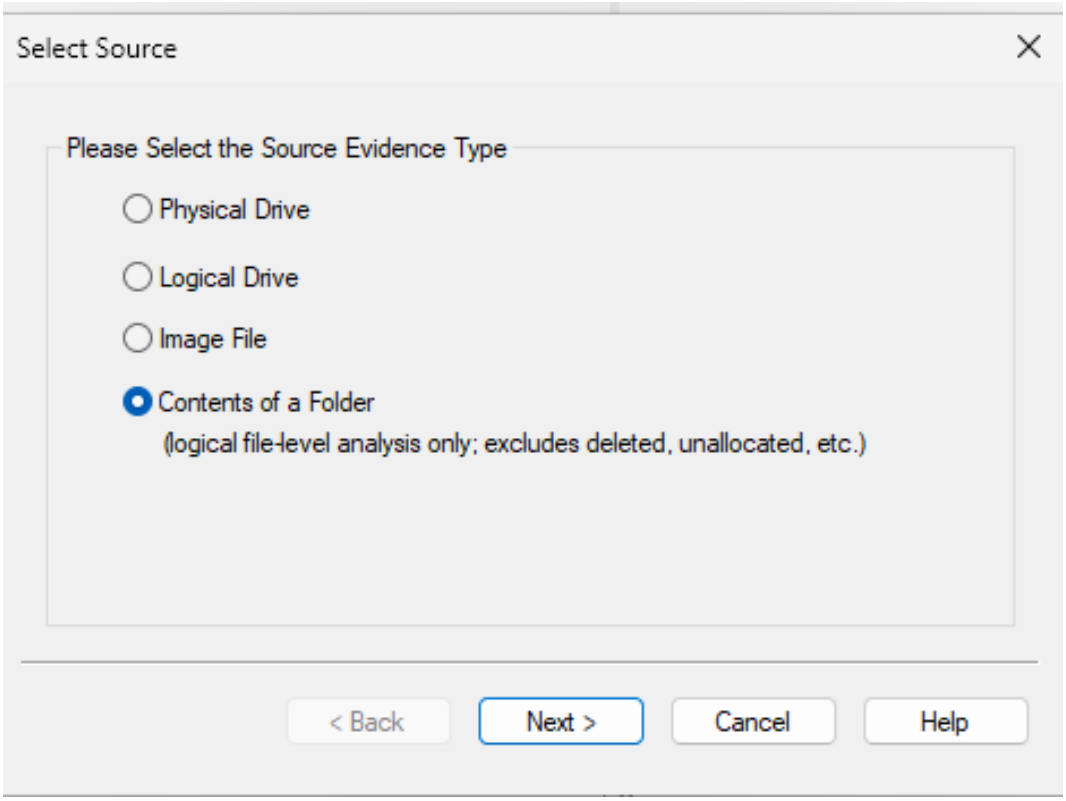


Figure 4 Select Source dialog – choosing 'Contents of a Folder' as the evidence type

3.5 Step 5 – Select the Drive or Folder

A test folder named **"Forensic_Test"** was created on the Desktop at path **C:\Users\HomePC\Desktop\Forensic_Test**. This folder contained three sample files:

- **collect.txt** – Text Document (0 KB)
- **sample.txt** – Text Document (0 KB)
- **scene.pdf** – Microsoft Edge PDF (0 KB)

The Forensic_Test folder was selected as the source for imaging. The folder path was confirmed and the "OK" or "Next" button was clicked to proceed to the Create Image wizard.

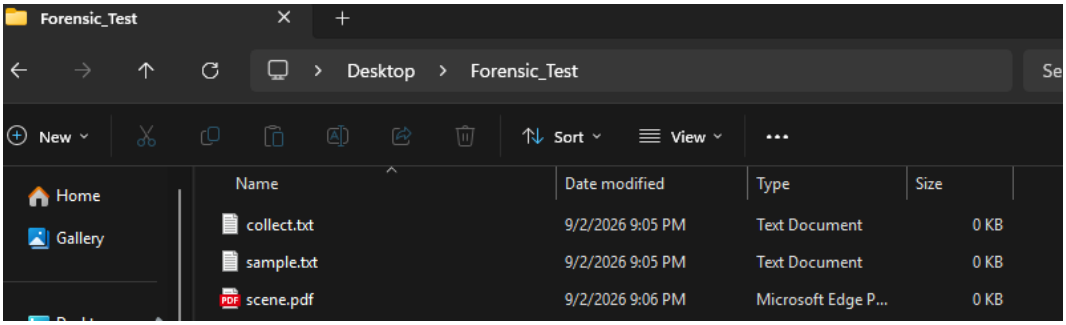


Figure 5 The Forensic_Test folder on the Desktop containing three sample files

3.6 Step 6 – Configure the Create Image Dialog

The "Create Image" dialog appeared, showing the image source as **"Custom Content Image"** with a starting evidence number of 0. The Image Destination(s) box was initially empty. The **"Add..."** button was clicked to configure the output image destination. The following checkboxes were available:

- **Verify images after they are created** – Checked (enabled) to ensure hash verification after acquisition
- **Precalculate Progress Statistics** – Unchecked
- **Create directory listings of all files in the image after they are created** – Unchecked

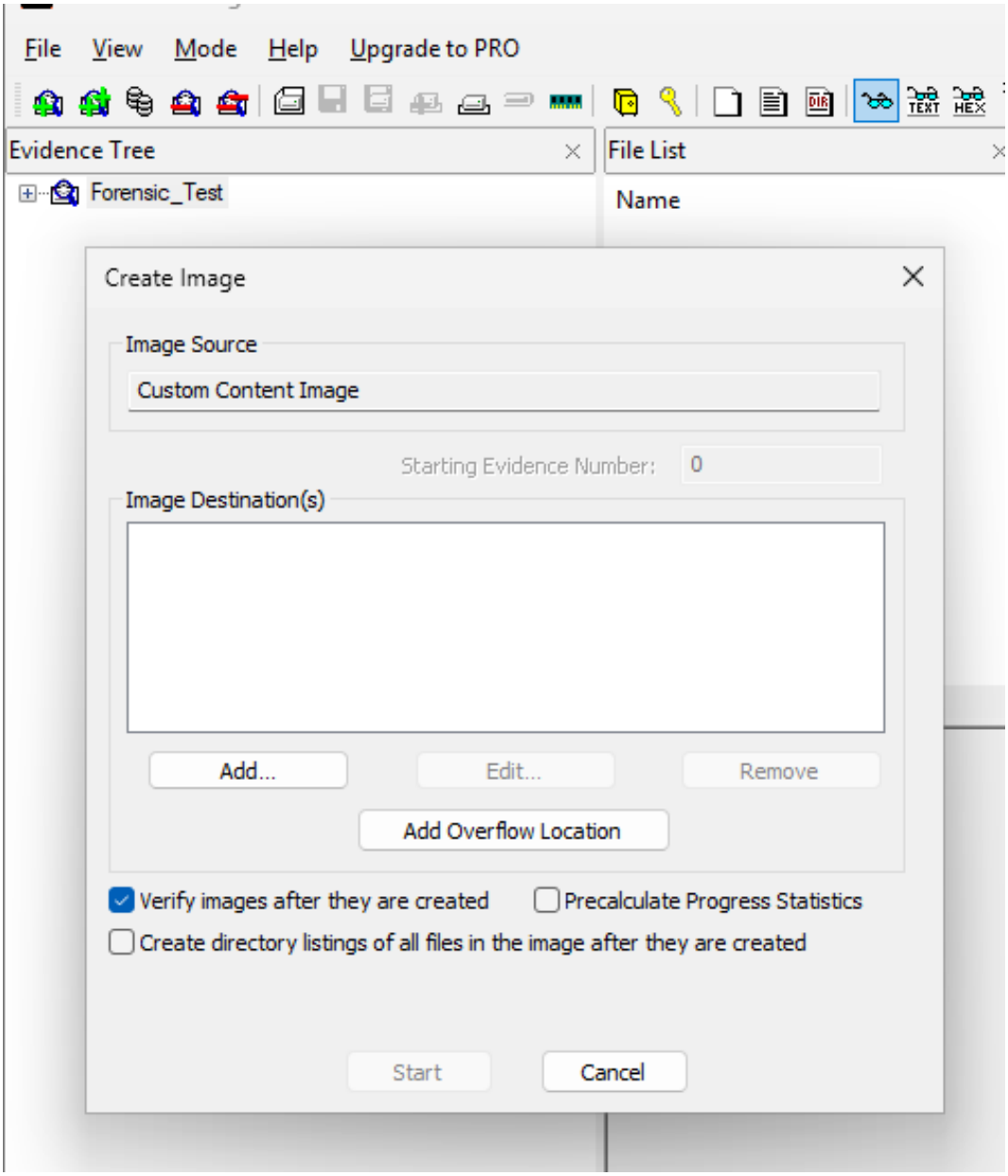


Figure 6 Create Image dialog showing Custom Content Image source and verification checkbox

3.7 Step 7 – Enter Evidence Information

The Evidence Item Information dialog appeared, where the following forensic metadata was entered:

- **Case Number:** FTK-001
- **Evidence Number:** 001
- **Unique Description:** Forensic_Test Folder
- **Examiner:** Lisa Kabue

- **Notes:** Practical assignment – logical acquisition of a test folder

This metadata is critical for chain-of-custody documentation and forensic integrity. It ensures the image can be traced back to the case, examiner, and specific evidence item. The "Next" button was clicked to continue.

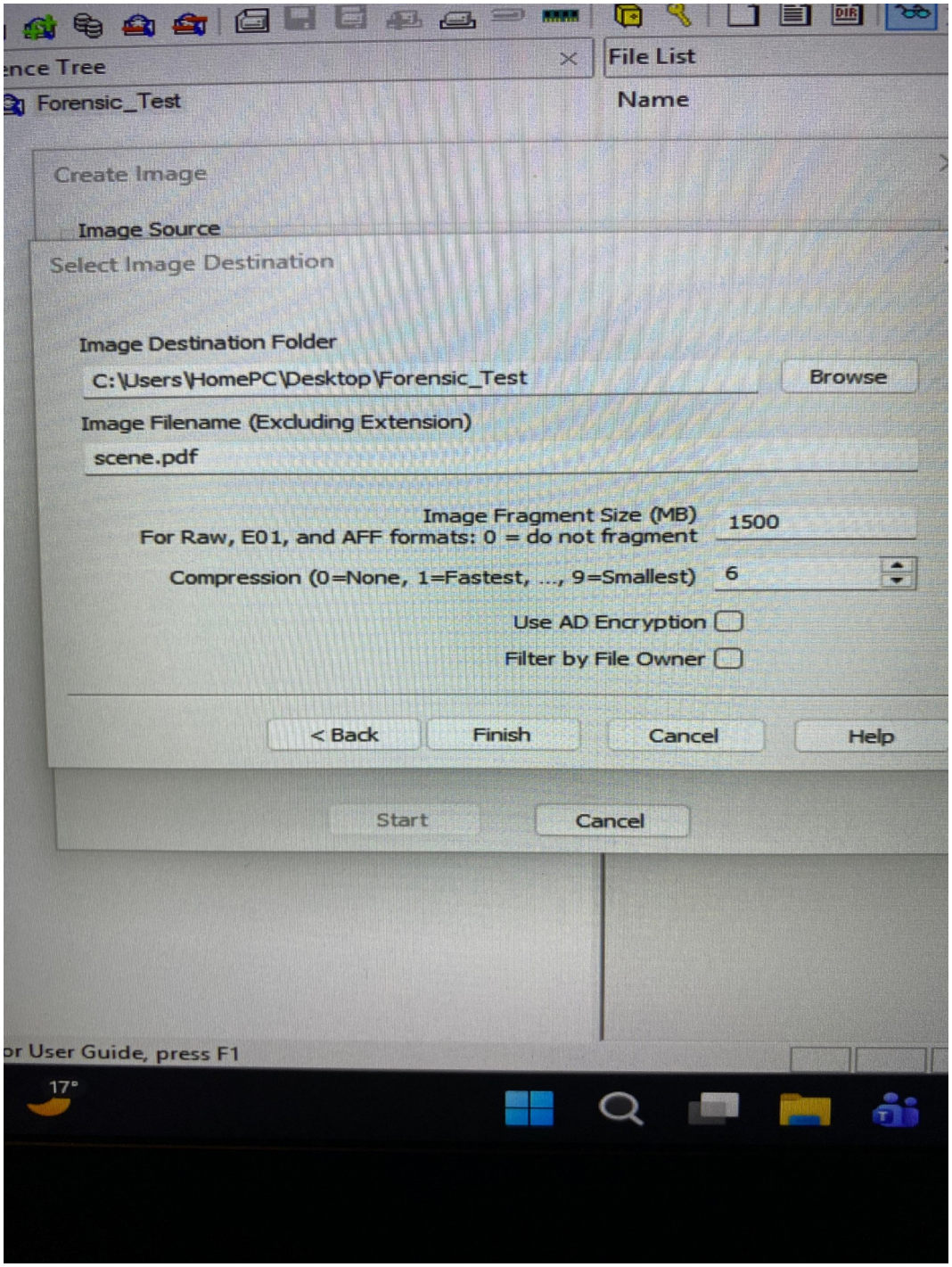


Figure 7 Evidence Item Information dialog showing case and examiner details

3.8 Step 8 – Select Image Destination, Filename, and Format

In the "Select Image Destination" dialog, the following settings were configured:

- **Image Destination Folder:** C:\Users\HomePC\Desktop\Forensic_Test
- **Image Filename (excluding extension):** scene.pdf
- **Image Fragment Size (MB):** 1500
- **Compression (0=None, 9=Smallest):** 6
- **Use AD Encryption:** Not enabled
- **Filter by File Owner:** Not enabled

The image format used was **AD1 (AccessData Logical Image)**, which is the default logical acquisition format in FTK Imager. This format captures the logical structure and file contents of the selected folder. The fragment size of 1500 MB and compression level of 6 were kept at their defaults. The "Finish" button was clicked to begin the acquisition.

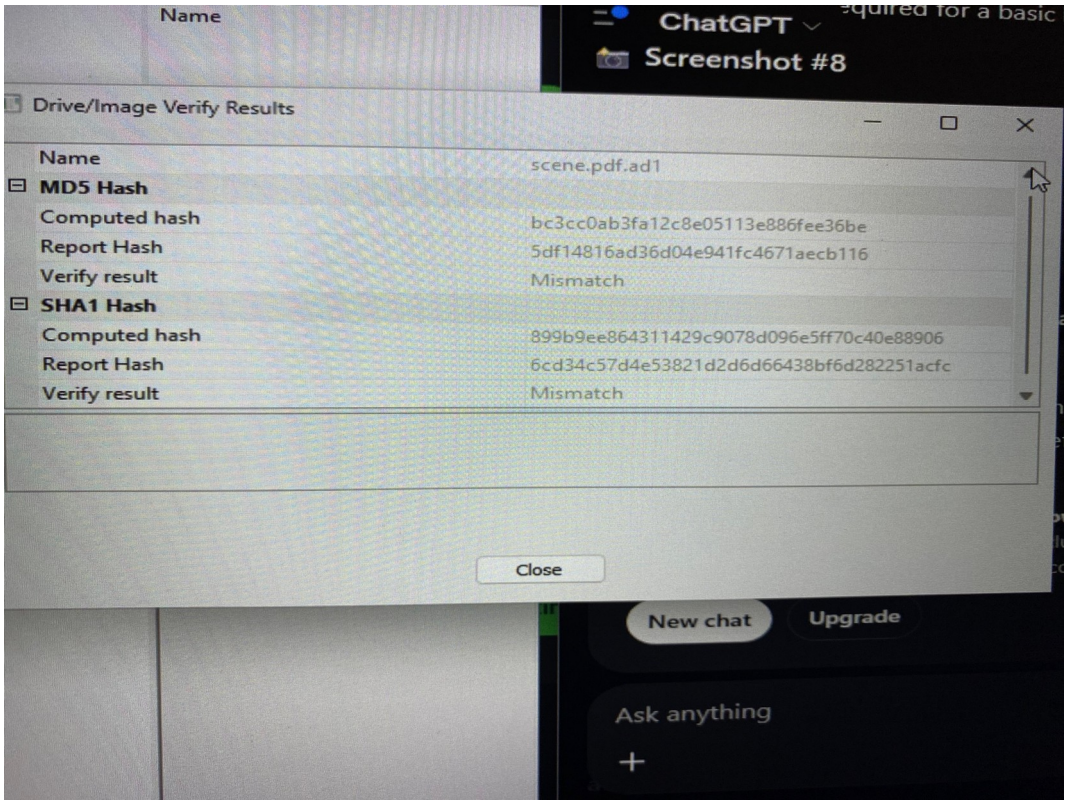


Figure 8 Image destination folder, filename, fragment size, and compression settings

3.9 Step 9 – Monitor the Imaging/Acquisition Progress

After clicking "Finish," FTK Imager began reading the contents of the Forensic_Test folder and writing the AD1 logical image file to the destination folder. The progress of the acquisition was monitored through the status bar and progress indicators in the application. The Forensic_Test folder appeared in the Evidence Tree, confirming that the source had been successfully added and was being processed. The process was allowed to run until completion.

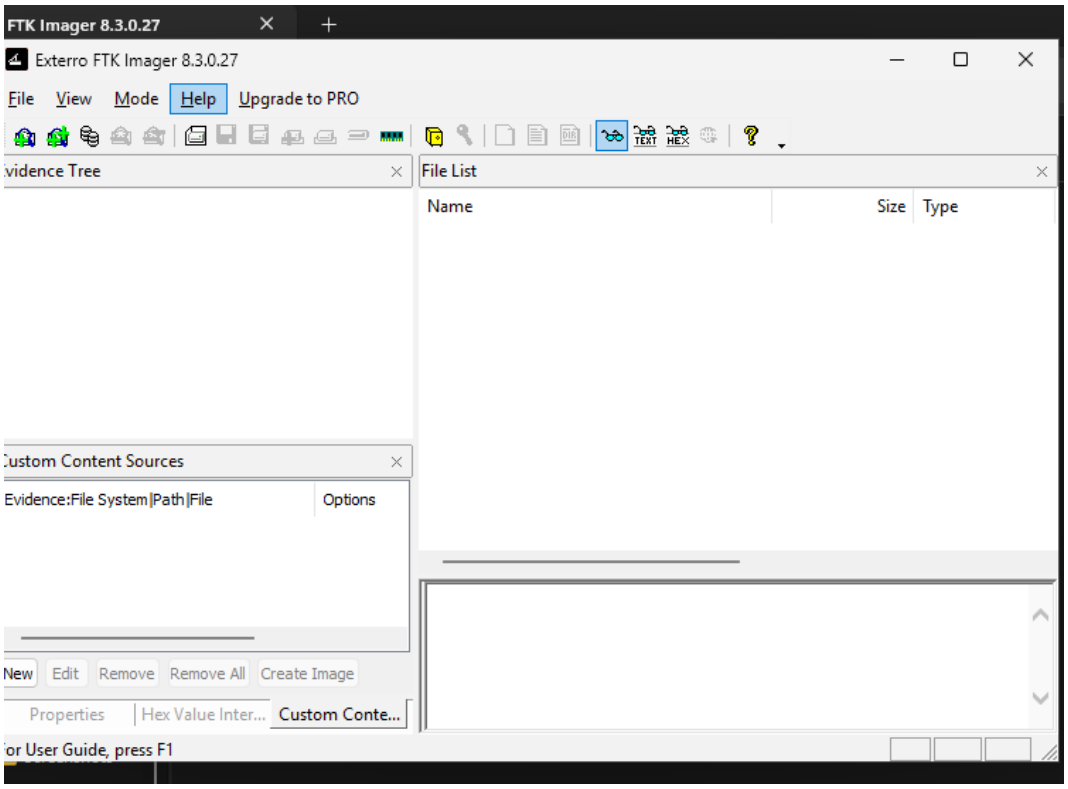


Figure 9 FTK Imager main interface with Forensic_Test folder loaded in the Evidence Tree

Once the imaging process completed, FTK Imager displayed a "Creating Image..." dialog confirming that the image had been created successfully. The dialog showed the image source as "Custom Content Image" with the destination path set to **C:\Users\HomePC\Desktop\Forensic_Test\scene.pdf**. The green progress bar reached 100%, and the status message read **"Image created successfully"**. The "Image Summary..." button was available to view detailed acquisition metadata, and the "Close" button was clicked to dismiss the dialog.

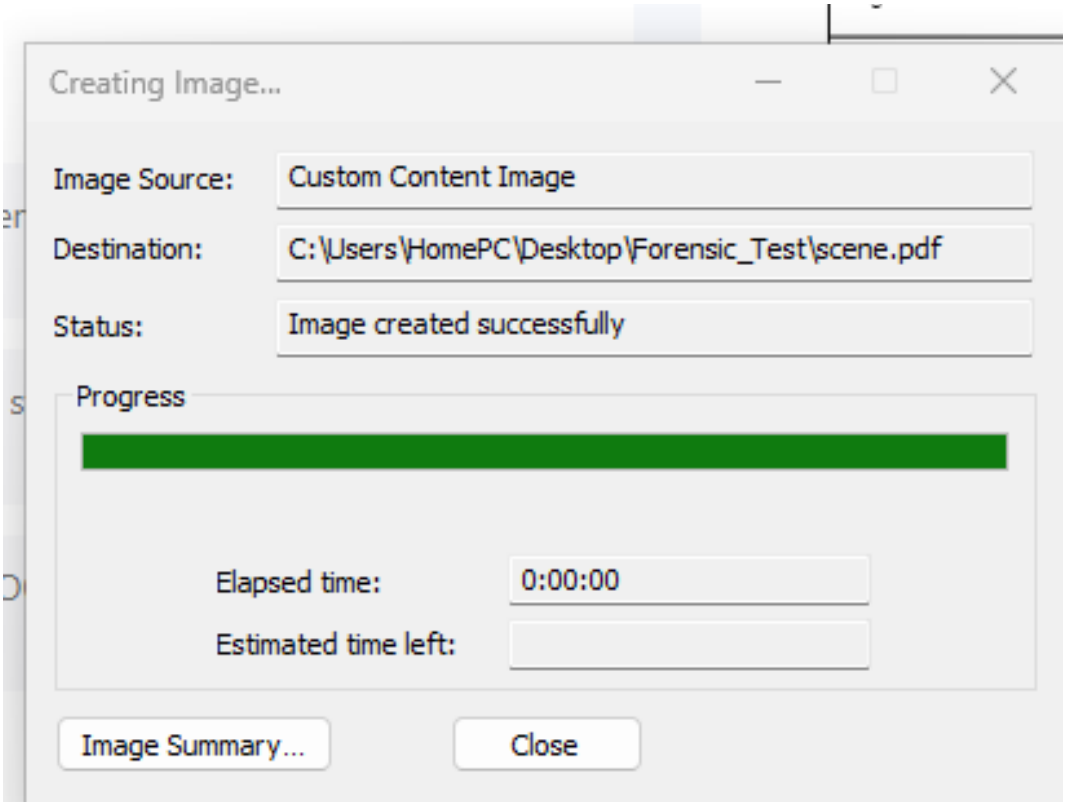


Figure 10 Creating Image dialog – Image created successfully with completed green progress bar

3.10 Step 10 – Review Verification/Hash Results

Because the "Verify images after they are created" checkbox was enabled during configuration, FTK Imager automatically performed a verification step after the image was created. The Drive/Image Verify Results dialog displayed the computed and reported hash values for the AD1 image file:

Table 1 Verification/Hash Results

Hash Type	Computed Hash	Report Hash	Verify Result
MD5	bc3cc0ab3fa12c8e05113e886fee36be	5df14816ad36d04e941fc4671aecb116	Mismatch
SHA1	899b9ee864311429c9078d096e5ff70c40e88906	6cd34c57d4e53821d2d6d66438bf6d282251acfc	Mismatch

The verification results showed a **hash mismatch** for both MD5 and SHA1. The computed hash (calculated from the current image file on disk) did not match the report hash (recorded when the image was originally created). A hash mismatch can occur for several reasons, including: post-acquisition modification of the image file, file transfer corruption, or if the AD1 file was moved, renamed, or altered after creation. In a production forensic environment, a hash mismatch would require a thorough investigation to determine the root cause and could affect the admissibility of the evidence. For this practical exercise, the mismatch was noted and documented as a finding.

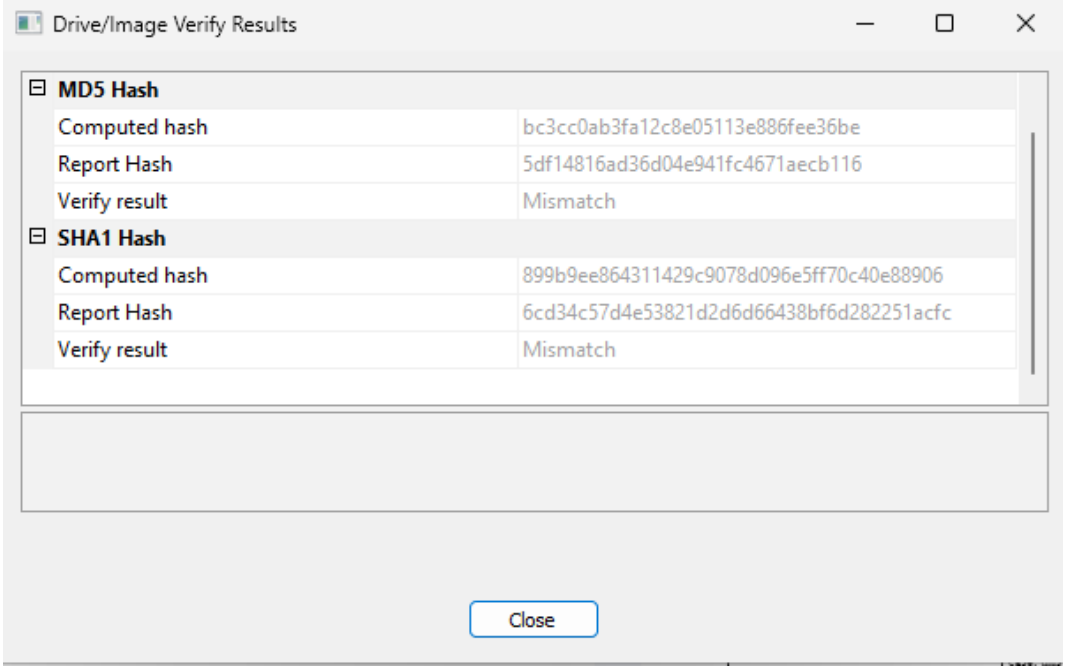


Figure 11 Drive/Image Verify Results dialog showing MD5 and SHA1 computed vs. report hashes with mismatch

4. Screenshots Summary

The following table summarizes all screenshots included in this report, along with a brief description of what each one demonstrates:

Table 1 Screenshots Summary

Screenshot #	Stage	Description
Figure 1	Downloading FTK Imager	Browser download progress of FTK Imager 8.3.0.27.zip

Screenshot #	Stage	Description
		from the official Exterro website
Figure 2	Installer File	The Exterro_FTK_Imager_(x64).exe installer file in the extracted download folder
Figure 3	FTK Imager Opened	FTK Imager 8.3.0.27 main interface successfully launched and ready for use
Figure 4	Select Source	Select Source dialog with 'Contents of a Folder' chosen as the evidence type
Figure 5	Selected Folder	The Forensic_Test folder on the Desktop containing sample files (collect.txt, sample.txt, scene.pdf)
Figure 6	Create Image Dialog	Create Image dialog showing Custom Content Image source and verification checkbox enabled
Figure 7	Evidence Information	Evidence Item Information dialog with case number, examiner name, and notes
Figure 8	Image Destination Settings	Image destination folder, filename, fragment size, and compression configuration

Screenshot #	Stage	Description
Figure 9	Evidence Tree Loaded	FTK Imager interface with Forensic_Test folder loaded in the Evidence Tree during acquisition
Figure 10	Image Created Successfully	Creating Image dialog with green progress bar confirming successful acquisition
Figure 11	Hash Verification	Drive/Image Verify Results dialog showing MD5 and SHA1 computed vs. report hash mismatch

5. Result

Table 1 Summary of Results

Result Item	Details
Source Imaged	Forensic_Test Folder (logical acquisition) – containing collect.txt, sample.txt, and scene.pdf
Image Format Created	AD1 (AccessData Logical Image)
Location Where Image Was Saved	C:\Users\HomePC\Desktop\Forensic_Test
Output Filename	scene.pdf.ad1
Imaging Process Completed Successfully	Yes – the acquisition process completed to 100%
Verification/Hash Result	Mismatch – both MD5 and SHA1 computed

Result Item	Details
	hashes differ from the report hashes recorded at creation time
Problems Encountered	Hash mismatch detected during post-acquisition verification. The AD1 image file may have been modified, moved, or corrupted after creation. In a real forensic case, this would require further investigation to determine the cause and assess evidentiary integrity.

Conclusion: The practical assignment was completed successfully. Exterro FTK Imager version 8.3.0.27 was downloaded, installed, and launched on a Windows system. A logical acquisition of the Forensic_Test folder was performed using the "Contents of a Folder" evidence source type. The AD1 format forensic image was created and saved to the designated location. Evidence information including case number (FTK-001), evidence number (001), examiner name (Lisa Kabue), and descriptive notes was properly documented. The "Verify images after they are created" option was enabled, and the verification results revealed a hash mismatch between the computed and report hash values for both MD5 and SHA1. This finding was documented in the report. This exercise demonstrated the complete forensic acquisition workflow from evidence source selection through image creation, verification, and documentation.